**MAHATMA GANDHI UNIVERSITY, KOTTAYAM**

**MGU-UGP (HONOURS) REGULAR EXAMINATION March 2026**

**Fourth SEMESTER**

**Discipline Specific Core Course (DSC) - MG4DSCCHE202 – FUNDAMENTALS OF PHYSICAL CHEMISTRY**

**(2024 ADMISSION ONWARDS)**

**Duration: 1 hour 30 minutes** **Maximum Marks: 50**

## Part A

**Multiple choice questions Answer all questions Each question carries 1 marks**

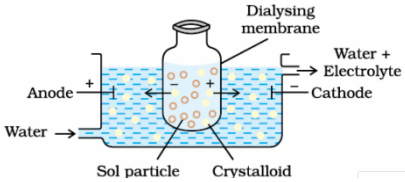
1. In a BCC unit cell, the atom at the body center contributes
   **b). 1**
2. Which among the following is a good adsorbent?
   **b). Charcoal**
3. Who invented galvanic cell?
   **b). Luigi Galvani and Alessandro Volta**
4. The forbidden energy gap is maximum in
   **c). Insulators**
5. Name the process in which the sol particles migrate toward the positive electrode during electrophoresis is called
   **b).anaphoresis**
6. Which statement is correct regarding the Kohlrausch's law?
   **b). Molar conductance at infinite dilution is the sum of individual ionic conductances**
7. Particles of suspension can be separated by
   **c). Ordinary filtration**
8. If the same current flows through $CuSO_4$ and $AgNO_3$ solutions, which will deposit more mass (Equivalent weight of Silver = 107.87 g; Copper = 31.77 g).
   **b). Silver**
9. Molar conductance (ΛM) is defined as the conductance of
   **d). solution containing 1 mole of electrolyte**
10. The electrochemical series is arranged according to:
    **b). Standard electrode potentials**

## Part B

**Short Answer Questions Answer any 4 questions Each question carries 5 marks**

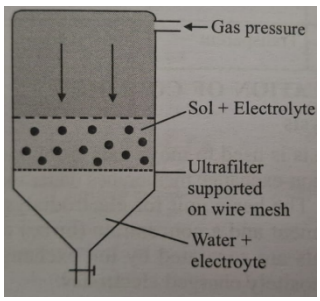
11. Explain how crystal systems are classified based on axial lengths and angles.
    **(Out of syllabus. Give full credit to all)**
12. Discuss briefly on a) electrodialysis b) ultrafiltration.

    Electrodialysis is used to move ions from one solution to another solution through ion-exchange membranes under the influence of electric field.




**(2.5 marks)**

Ultrafiltration is the process of separating the colloidal particles from the solvent and soluble solutes present in the colloidal solution by specially prepared filters, which are permeable to all substances except the colloidal particles.

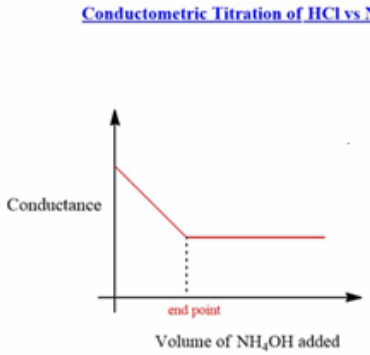



**(2.5 marks)**

13. Mention any four applications of conductance measurements
    (a) Coloured solution can be titrated where indicator cannot be used
    (b) A mixture of strong and weak acids can be titrated
    (c) Accurate results can be obtained as end point is determined graphically
    (d) Weak acid weak base titration can be carried out
    (e) No special care is required for end point determination

    **(Give full credit to any relevant four applications)**
14. What are temporary magnets? Why is soft iron preferred for making temporary magnets?
    A magnet does not retain its magnetic property once the magnetic field is removed. Soft iron is preferred for temporary magnets because it gets magnetized easily (high permeability) and loses its magnetism quickly (low retentivity and coercivity) when the external magnetic field is removed.

    **(Definition with reasonable explanation 5 marks)**
15. Discuss the nature of graph obtained in the conductometric titration of strong acid versus weak base by taking suitable examples.
    In the conductometric titration of a strong acid with a weak base (e.g., HCl vs $NH_4OH$), the graph shows a sharp decrease in conductance and after the end point the graph becomes almost horizontal.




**(Graph 3 marks, explanation with example 2 marks)**

16. Define conductors. Explain the effect of temperature on the conductivity of conductors.
    Substances that allow electric current to pass through them are known as conductors. For metallic conductors, as temperature increases conductivity decreases. For electrolytic conductors, as temperature increases conductivity increases.

    **(Definition 2.5 marks, explanation 2.5 marks)**

## Part C

## Short Essay Questions Answer any 2 questions Each question carries 10 marks

17. What are emulsions? Discuss the applications of emulsions in medicine and cleansing action of soap.

An emulsion is a colloidal system of two immiscible liquids in which one liquid acts as the dispersed phase while the other liquid acts as the dispersion medium. Two types Oil in water type and water in oil type.

**Medicines**

In medicines Colloidal system is used as therapeutic agents in different areas. Example: -Silver colloids as germicidal agents

Copper colloids as anticancer drugs. (i) Colloidal dispersion of gelatin is used in coating over tablets and granules which upon drying leaves a uniform dry film over them and protect them from adverse conditions of the atmosphere.

Targeted drug delivery: Targeted drug delivery system is a special form of drug delivery system where the medicament is selectively targeted or delivered only to its site of action or absorption and not to the non-target organs or tissues or cells. Colloidal systems are used as the targeted drug delivery system.

**Cleansing action of soap**

The way soap removes dirt and grease is a classic example of emulsification. Dirt is usually oily, and since water and oil don't mix, water alone can't be used to clean. A soap molecule has two distinct ends: Hydrophilic Head: Water-loving and polar and Hydrophobic Tail: Water-fearing and non-polar (usually a long hydrocarbon chain). When we apply soap to greasy hands, the hydrophobic tails "dive" into the oil/grease, while the hydrophilic heads stay pointing out toward the water. The soap molecules surround the oil droplet, forming a spherical structure called a micelle. The oil is trapped in the centre. As we scrub, emulsification happens and these micelles are lifted off the surface. Because the outer layer of every micelle is negatively charged (the heads), they repel each other. This prevents the oil droplets from clumping back together.

**(Definition 2 marks, Application in medicine - 3 marks, Cleansing action of soap- 5 marks)**

18. Write a short note of the following a) Hardy-Schulz rule b) electrical double layer

(a) The greater the valency of the flocculating ion added, the greater is its power to cause precipitation. This is known as Hardy-Schulze rule.

The coagulating power increases with the valency of an ion.

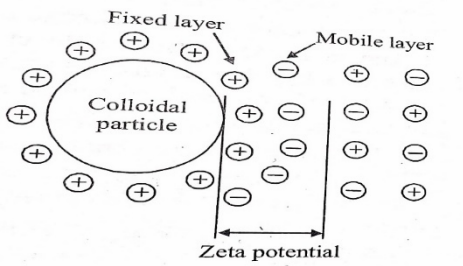
For example, in the coagulation of a negative sol, the order of flocculating power can be depicted as:

$Al^{3+} > Ba^{2+} > Na^{+}$

Similarly, in the coagulation of a positive sol, the order of flocculating power can be depicted as:

**(5 marks)**

(b) The electrical double layer is a structure that forms at the interface between a charged surface (such as a colloidal particle or electrode) and a surrounding liquid containing ions. It consists of two layers of opposite charges, which play a crucial role in the stability and behaviour of colloidal systems. The first layer is fixed (Stern layer) and the second layer is mobile which is called diffused layer (Gouy- Chapmann layer). Potential difference of this electrical double layer is called Zeta potential(Z).




**(5 marks)**

19. Explain the principle and working of redox electrodes using a platinum electrode.

An oxidation-reduction (redox) electrode is an electrical conductor used in an electrochemical cell where simultaneous oxidation and reduction reactions occur, involving the transfer of electrons. Here an inert metal is dissolved in a solution containing both the oxidized and reduced form of a molecule or ion. The inert metal simply provides electrical contact

Eg; Pt I $Fe^{2+}$, $Fe^{3+}$; Pt I $Sn^{2+}$, $Sn^{4+}$

Eg: Quinhydrone electrode

**(Reasonable explanation about redox electrode, give full credit)**

20. Explain the principle, instrumentation, and working of potentiometric titration. Discuss the role of indicator electrode and reference electrode with suitable examples

Potentiometric titration is a volumetric method in which the potential difference between two electrodes is measured as a function of the added titrant volume. The principle rests on the Nernst Equation, which relates the electrode potential to the concentration of the species in the solution:

$$E = E^\circ + \frac{RT}{nF} \ln \frac{[\mathrm{Ox}]}{[\mathrm{Red}]}$$

**Instrumentation**- 3 main components

- **Reference Electrode**: An electrode with a constant, known potential (e.g., Saturated Calomel Electrode or Ag/AgCl electrode).
- **Indicator Electrode**: An electrode whose potential depends on the concentration of the analyte (e.g., Glass electrode for pH, Platinum wire for redox).
- **Potentiometer/pH Meter**: To measure the potential difference between the two electrodes**.**

**Working**

- The analyte solution is taken in a beaker.
- The indicator electrode and reference electrode are immersed in the solution.
- The titrant is added gradually from the burette.
- After each addition, the potential difference (E) is measured.
- A graph is plotted between: Potential (E) vs Volume of titrant, or ΔE/ΔV vs Volume (for sharper endpoint detection).
- The equivalence point is identified from the sharp jump in potential or peak in derivative curve.

Indicator electrode responds to the activity (concentration) of the analyte ion and reference electrode maintains a constant potential regardless of solution composition. For example, in case of acid- base titration indicator electrode is Glass electrode and reference electrode is Calomel electrode.

**(Principle- 2 marks, Instrumentation- 3 marks, working - 3 marks, Electrodes explanation- 2 marks)**